

Name: _____ Tutorial group: _____

Matriculation number:

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20 September 2020 **MH1812 Test 1** 60 minutes

QUESTION 1. (30 marks)

Use mathematical induction to show that

$$1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1}n^2 = (-1)^{n-1} \cdot \frac{n(n+1)}{2}$$

whenever n is a positive integer.

[Proof:]

Base case: when $n = 1$, the left-hand side is $1^2 = 1$ and the right-hand side is $(-1)^0 \cdot \frac{1 \cdot 2}{2} = 1$. The equality holds.

Inductive step: Suppose that the equality holds for $n = k$, and we shall show that it holds for $n = k + 1$, that is, we shall show that

$$1^2 - 2^2 + 3^2 - \dots + (-1)^{k-1}k^2 + (-1)^k(k+1)^2 = (-1)^k \frac{(k+1)(k+2)}{2}. \quad (1)$$

By induction hypothesis, the LHS of (1) equals

$$(-1)^{k-1} \cdot \frac{k(k+1)}{2} + (-1)^k(k+1)^2 = (-1)^k(k+1) \cdot \left(-\frac{k}{2} + k + 1\right) = (-1)^k(k+1) \frac{k+2}{2},$$

which is exactly the RHS of (1). This completes the proof of the inductive step.

Therefore, by mathematical induction, the original equality holds for all integers $n \geq 1$.

For graders only	Question	1	2(a)	2(b)	3(a)	3(b)	Bonus	Total
	Marks							

QUESTION 2. (30 marks)

(a) (10 points) What is $2020^{1812} \bmod 30$?

[Solution:] First, we have $2020 \bmod 30 = 10$. Observe that $10^2 \bmod 30 = 10$, we know that $10^n \bmod 30 = 10$ for all $n \geq 1$. Therefore $2020^{1812} \bmod 30 = 10^{1812} \bmod 30 = 10$.

(b) Let $\mathbb{R}$ denote the set of reals. For $x, y \in \mathbb{R}$, let $P(x, y)$ denote the predicate “ $x^2 - x + 2020y \geq 0$ ”. What are the truth values of these statements?

(i) (10 points) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, P(x, y)$.

(ii) (10 points) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, P(x, y)$.

[Solution:]

(i) The statement is true. One can just choose $y = -(x^2 - x)/2020$.

(ii) The statement is false. For any $x \in \mathbb{R}$, let $y = -(x^2 - x)/2020 - 1$, then $P(x, y)$ is false.

QUESTION 3.

(40 marks)

(a) (20 points) Show that $q \wedge \neg(p \rightarrow q)$ is a contradiction.

[Proof:]

$$\begin{aligned} q \wedge \neg(p \rightarrow q) &\equiv q \wedge \neg(\neg p \vee q) \\ &\equiv q \wedge (p \wedge \neg q) \\ &\equiv (q \wedge \neg q) \wedge p \\ &\equiv F \wedge p \\ &\equiv F \end{aligned}$$

(b) (20 points) Determine whether the following argument is valid¹.

$$\begin{aligned} &(p \wedge q) \rightarrow (r \vee s); \\ &\neg r; \\ &p \rightarrow q; \\ &p; \\ &\therefore s. \end{aligned}$$

[Solution:] The argument is valid, as shown by the following inference table.

Step	Formula	Reason
(1)	$p \rightarrow q$	Premise
(2)	p	Premise
(3)	q	(1)+(2), modus ponens
(4)	$p \wedge q$	(2)+(3), conjunctive addition
(5)	$(p \wedge q) \rightarrow (r \vee s)$	Premise
(6)	$r \vee s$	(4)+(5), modus ponens
(7)	$\neg r$	Premise
(8)	s	(6)+(7), disjunctive syllogism

BONUS QUESTION.

(10 marks)

[Points will be given to *fully* correct solutions only. The total mark of this test is capped at 100 marks.]

Prove that there are no integers x and y such that $x^2 + y^2 = 444444443$.

[Proof:] Note that $n^2 \equiv 0, 1 \pmod{4}$ for any integer n (this can be readily verified by calculating $n^2 \pmod{4}$ for $n = 0, 1, 2, 3$). Observe that the right-hand side is congruent to 3 modulo 4, therefore it cannot be written as a sum of two squares.

¹The inference rules you may use are: modus ponens, modus tollens, conjunctive simplification, conjunctive addition, disjunctive addition, disjunctive syllogism, rule of contradiction and disjunction elimination.